Mahmoud Alyosify

AI Engineer, Educator & Community Volunteer

New Cairo, Egypt · Open to relocation and remote · +20 114 555 7959 · mahmoud.alyosify@gmail.com
Portfolio: mahmoudalyosifysite.github.io · linkedin.com/in/mahmoudalyosify · huggingface.co/mahmoudalyosify

PROFILE

Egyptian AI engineer and educator who has spent six years **sharing knowledge**: free and low-cost Arabic computer-science courses for 10,000+ learners, student leadership at Assiut University, and part-time teaching at NTI. Graduated in October 2026 with an MSc in Artificial Intelligence from Queen's University, Canada (GPA 4.30 / 4.30) on a competitive presidential scholarship. Focused on using AI for problems that matter to people: health, safety and access to education.

VOLUNTEERING & LEADERSHIP

Volunteer Team Leader
USAID University Center for Career Development — Assiut, Egypt

- Led a volunteer team helping university students prepare for their first jobs; coordinated activities and mentored peers.

Secretary, Higher Scientific & Technological Committee
Student Union, Assiut University — Assiut, Egypt

- Elected to organise the faculty's scientific and technology events and competitions.

Student Ambassador — 2022 – 2023
Information Technology Institute (ITI) — Assiut, Egypt

- Guided students to ITI's free national technical-training programs.

Open Educator, Arabic Computer Science — 2020 – Present
Udemy & YouTube — Remote

- Arabic courses in algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.
- Won 1st Place in a university Science & Technology Content Competition.

TEACHING & PROFESSIONAL EXPERIENCE

Part-Time Machine Learning Instructor — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour hands-on machine learning bootcamps**: regression, SVM, KNN, Naive Bayes, PCA and neural networks, in Python labs.
- Teach the **AI for Business** module of the same program, on a separate engagement.
- Write the final exams and assess the trainees' end-to-end capstone projects.

AI Engineer Intern — Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) — Maadi, Cairo

- Building a RAG assistant that answers employees' questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.

AI in Cybersecurity Intern — Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) — Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) — Feb 2024 – Mar 2025
National Company for Roads Building & Development — Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.

PROJECTS FOR PEOPLE

Vitalism Solution — graduation project — 2023
Python, computer vision, signal processing (rPPG)

- Measures heart rate and other vital signs from ordinary video, with no contact sensors. Grade **A+**; **Microsoft Imagine Cup 2023 semi-finalist**.

- Led a six-student team from idea to a working prototype; Finalist at ISEIC 2023.

MinimalLM — Master's thesis, published on arXiv 2026
Python, PyTorch, Hugging Face Transformers, NVIDIA A40

- MSc thesis: a training-free method that makes open LLMs generate **47.6% fewer tokens** and answer 19–68% faster, with no answer cut off.

AudioShield — deepfake audio detection
Python, CNN, Streamlit

- Trained a CNN to detect deepfake (synthetic) speech for voice-fraud and digital-forensics use; live predictions in a Streamlit app.

SCRAP — satellite collision risk prediction 2026
Python, XGBoost, LightGBM, Optuna, SHAP

- Predicted satellite collision risk using only data available **48+ hours** before closest approach, so the warning arrives in time to act.

PUBLICATIONS

Say Less, Mean Better: Training-Free Verbosity Control for Language Models
M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026

Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding
M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

RECOGNITION

- Digilians Presidential Scholarship for graduate study in Canada (2025)
- Semi-Finalist, Microsoft Imagine Cup 2023 · Finalist, ISEIC 2023
- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Ideal Student Award, Assiut University (2022–2023)

SKILLS

People: team leadership, volunteer coordination, mentoring, public speaking, teaching in Arabic and English
Technical: artificial intelligence, machine learning, LLMs, Python, data analysis, cloud (AWS)

LANGUAGES

Arabic (native) · English (professional working proficiency)